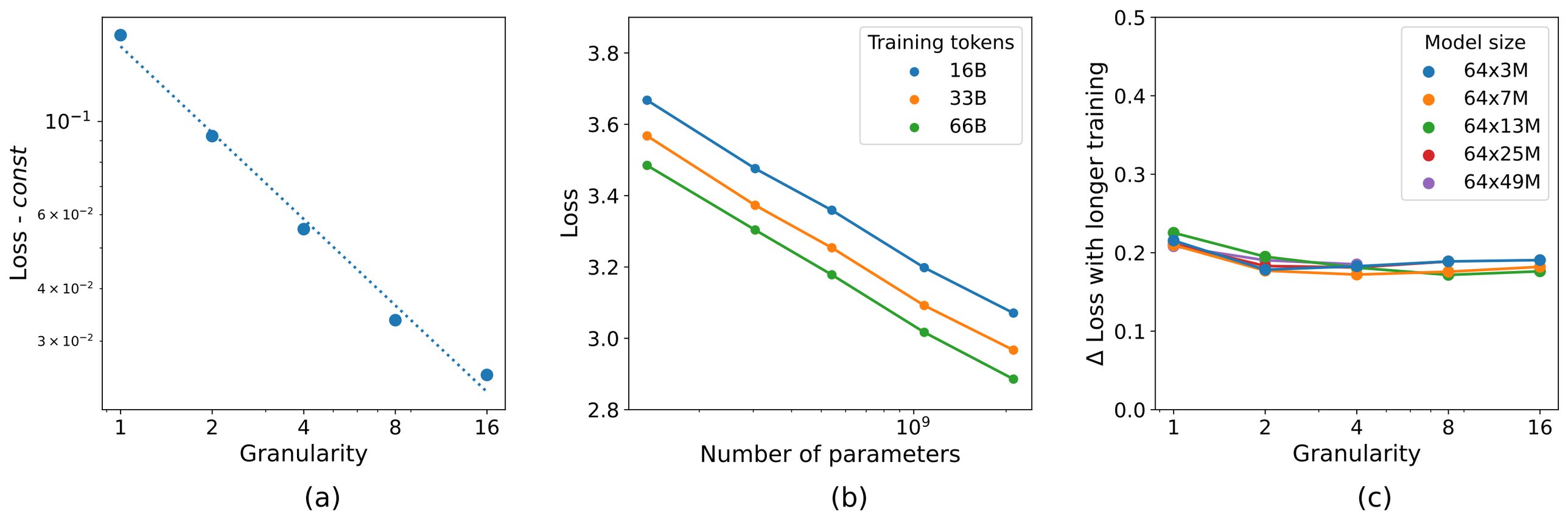
Loss - const
10^{-1}
6×10^{-2}
4×10^{-2}
3×10^{-2}
1
2
4
8
16
Granularity
(a)
Loss
3.8
3.6
3.4
3.2
3.0
2.8
10^9
Number of parameters
Training tokens
16B
33B
66B
(b)
Δ Loss with longer training
0.5
0.4
0.3
0.2
0.1
0.0
1
2
4
8
16
Granularity
Model size
64x3M
64x7M
64x13M
64x25M
64x49M
(c)